



Budget-Constrained Online Bidding

Learning to bid in first-price auctions under a shared budget — from a single stochastic campaign to non-stationary multi-campaign settings.

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Requirements 1 – 4 · lab-aligned baselines and targeted extensions

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Problem Setting

First-price auction, each round

$$\text{win} \iff b_{i,t} \geq m_{i,t}$$

$m_{i,t}$ = highest competing bid for campaign i — the only uncertainty in the auction.

Utility and cost on a win

$$f = v_i - b_{i,t}, \quad c = b_{i,t}$$

Higher bids win more often but earn less per win and spend more budget.

Global budget constraint

$$\sum_t c_t \leq B, \quad \rho = B/T$$

ρ is the per-round spend rate every learner has to respect on average.

CONFLICT GRAPH

Some campaigns are incompatible: connected campaigns cannot both bid in the same round. With multiple campaigns the action becomes a **super-action**: a feasible combination of bids, one for each campaign, selected at each round.

Across the four requirements:

1 → 4 campaigns

stochastic → non-stationary

semi-bandit → full feedback

Working Method

STEP 1

Lab-aligned baseline

Course tools only — UCB, LP over distributions, Hedge, OGD — with the constants seen in the labs.

STEP 2

Baseline Behavior

Read behaviour over time — regret, spend, chosen bids — and locate where the baseline is conservative.

STEP 3

Targeted extension

Minimal change aimed at the observed defect. Out-of-course methods are explored and analyzed.

STEP 4

Ablation

Same environment, same seeds, one change at a time — so every gain is attributed to its cause.

Every comparison is paired: baseline and improved learner run on the same realizations of the environment. Differences in the curves are differences between algorithms, not between random draws.

REQUIREMENT 1

Single Campaign, Stochastic Environment

Two required learners: budget-blind UCB1, and a UCB-like learner with the budget constraint handled by a linear program.

$T = 5000$

$B = 125 \rightarrow \rho = 0.025$

$v = 0.6$

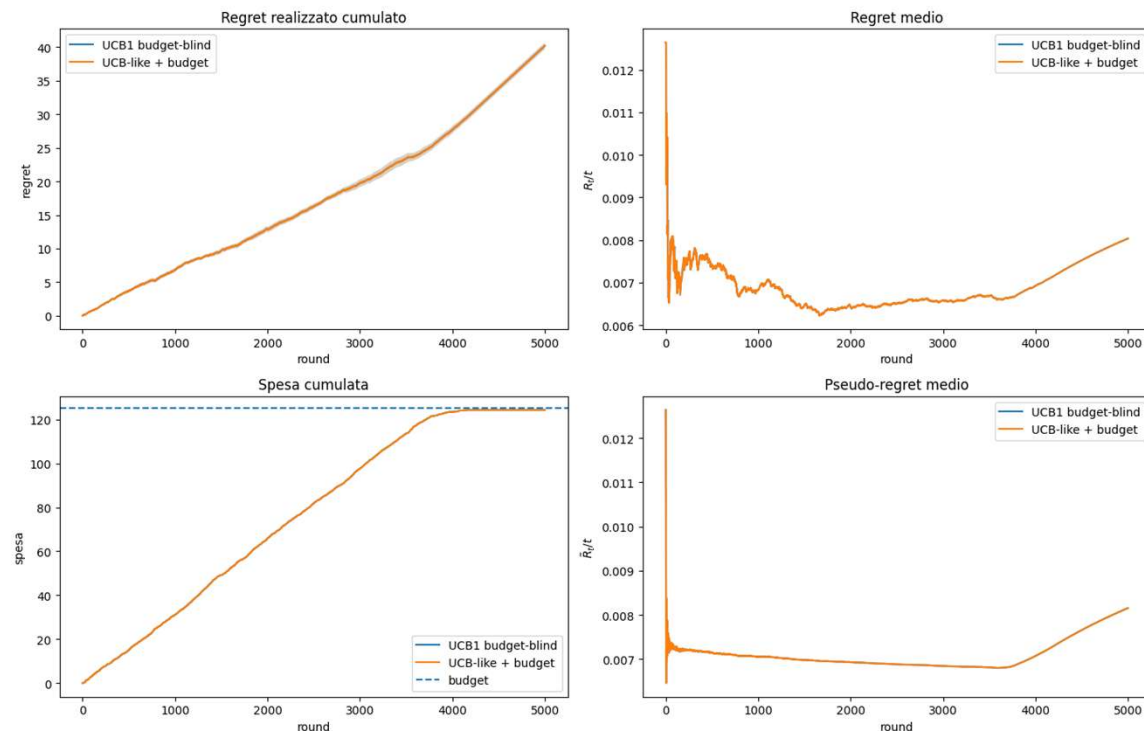
3 uniform competitors $\rightarrow P(\text{win}) = b^3$

discrete bid grid, $b \leq v$

BENCHMARK

Stochastic clairvoyant: LP over bid distributions with true win probabilities, spend $\leq \rho$. On this instance it mixes 96.6% on bid 0.4 — expected reward ≈ 0.0126 per round.

Baseline: the Two Learners Are Nearly Identical



40.2

final regret, UCB1
budget-blind

39.6

final regret, UCB-like +
budget LP

Why they coincide?

Hoeffding intervals stay wide over the whole horizon; the cost LCB is truncated at 0, so the LP constraint is almost never informative.

Both exhaust the budget early — the null bid is played in $\approx \frac{1}{3}$ of the rounds — and reach only **~36%** of the clairvoyant reward.

The bottleneck is **budget handling**, not reward estimation.

Budgeted KL-UCB — Three Changes to the Learner

1 · ONE ESTIMATE: WIN PROBABILITY

$$\hat{r} = (v - b) \hat{p}, \quad \hat{c} = b \hat{p}$$

At a fixed bid, reward and cost are both driven by the win/loss outcome. We estimate the win probability directly and derive both from it.

2 · KL CONFIDENCE BOUND

$$U_t = \max\{q \in [\hat{p}, 1] : N \text{kl}(\hat{p}, q) \leq \log t\}$$

Replaces Hoeffding's $\hat{p} \pm \sqrt{(\log t) / (2N)}$ with a KL-based bound, introduced to get tighter confidence estimates.

3 · AFFORDABILITY FILTER

$$b \text{ playable} \iff b \leq B_t$$

A bid enters the LP only if the remaining budget covers it; bid 0 stays as the zero-cost opt-out.

THE $\text{kl}(\cdot, \cdot)$ FUNCTION $\text{kl}(p, q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q}$

KL is a **Bernoulli-specific distance** between two win rates. It gives tighter bounds near 0 and 1, where Hoeffding tends to be too conservative.

Relevant win probabilities are tiny — $P(\text{win}) = b^3$ is **0.027** at $b = 0.3$, **0.064** at $b = 0.4$. Near 0, Hoeffding's radius $\sqrt{(\log t) / (2N)}$ is far wider than needed: every p_{UCB} saturates at 1 and the optimistic reward $(v - b) \cdot p_{\text{UCB}}$ ranks bids **backwards**. The KL interval exploits the Bernoulli variance vanishing near 0 and tightens exactly where it matters.

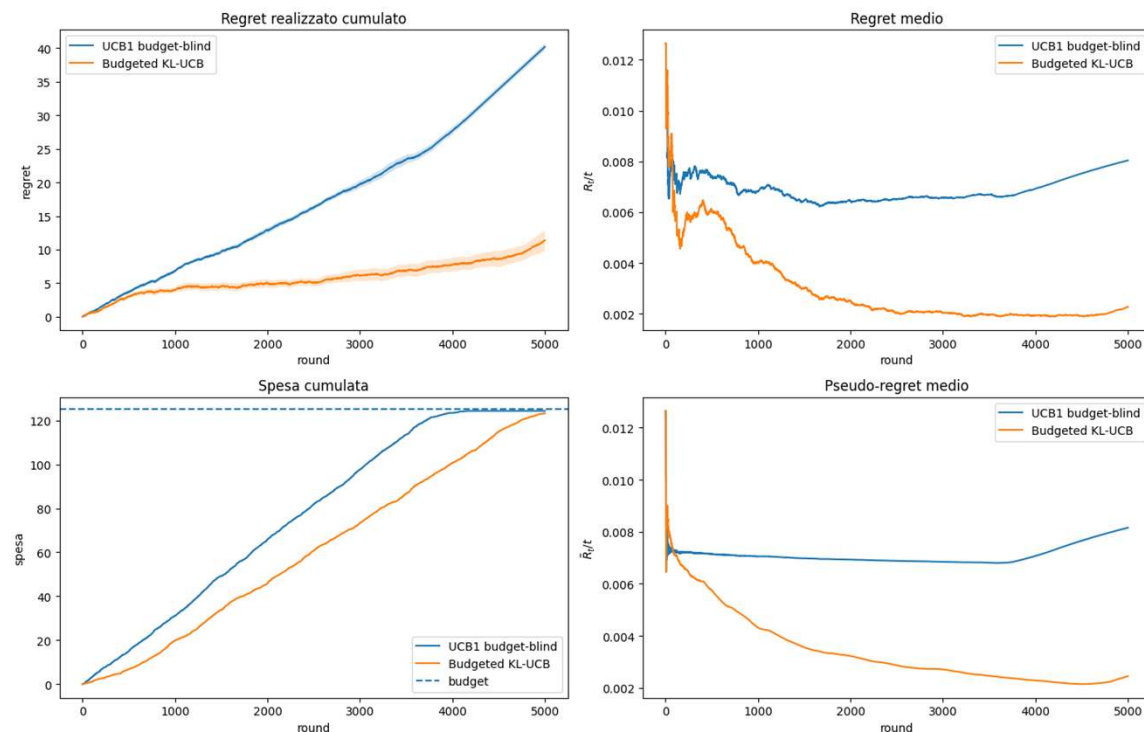
Unchanged:

same LP, $\rho = B/T$ fixed

same environment & seeds

dynamic pacing tried → no effect here, removed

Results and Ablation



40.2 → **11.4** final regret

82%

of clairvoyant reward
(was 36%)

123 / 125

budget spent, paced to
the horizon

ABLATION — SAME SEEDS, ONLY THE BOUND CHANGES

KL **11.4** (spend 123/125) vs Hoeffding **35.7**

(spend 45/125): Hoeffding keeps insisting on low bids, rarely wins, and cannot even spend the budget. The gain comes from the bound.

Multiple Campaigns, Combinatorial-UCB with Budget

One super-action per round — a feasible bid profile over the conflict graph — with semi-bandit feedback and a shared budget.

4 campaigns · $v = (0.95, 0.85, 0.78, 0.90)$

$T = 3000$ · $B = 700 \rightarrow \rho \approx 0.233$

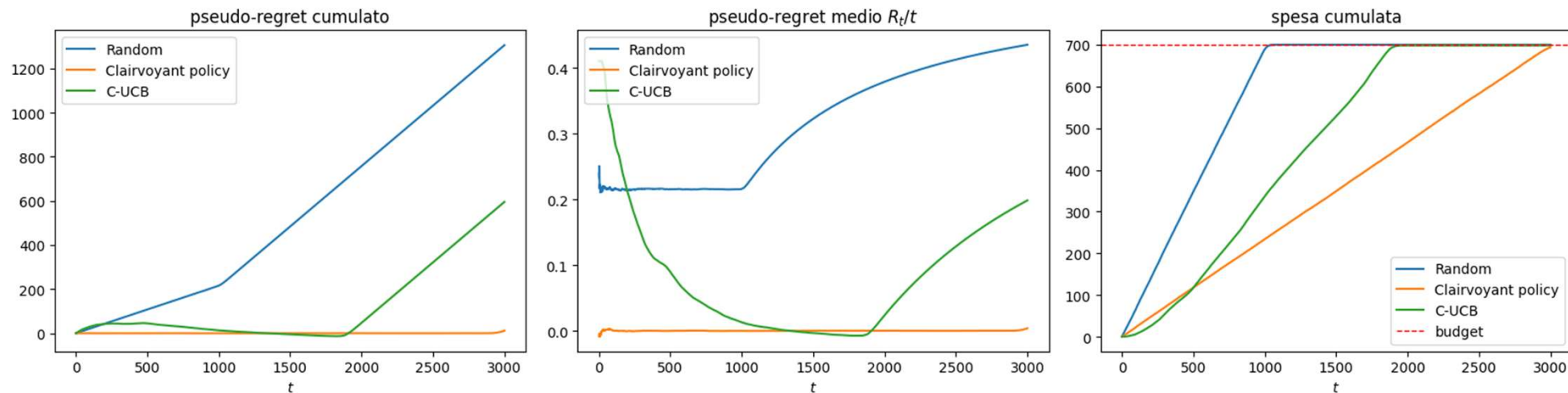
conflict chain 0-1-2-3

33 atomic arms · 250 super-actions

Beta marginals + 30% common factor $\rightarrow$ corr ≈ 0.3

BENCHMARK Clairvoyant LP over feasible super-actions with true expected reward and cost, under the average budget rate $\rho = B/T$.

Baseline: Learning Works, Pacing Fails



MID-HORIZON

Pseudo-regret ≈ 0 — indistinguishable from the clairvoyant policy: **estimation is fine**.

$t \approx 1900$

Cost LCB underestimates true spend $\rightarrow$ budget exhausted early $\rightarrow$ only the null action remains, regret restarts **linear**.

FINAL

Pseudo-regret ≈ 595 (0.198/round) vs random 0.435/round. Budget never violated (max 699/700).

Four Targeted Changes to the Learner

1 · $p(\text{win})$ PER ARM

$$\hat{r} = (v_i - b) \hat{p}, \quad \hat{c} = b \hat{p}$$

One Bernoulli estimate per (campaign, bid) — semi-bandit feedback makes it valid arm by arm.

2 · KL BOUND ON $p(\text{win})$

$$\max\{q \geq \hat{p} : N \text{kl}(\hat{p}, q) \leq \log t\}$$

Same motivation as R1: many arms have small win probabilities where Hoeffding remains too loose and ranks arms less accurately.

3 · EMPIRICAL COST IN THE LP

$$\hat{c}(a) = b \hat{p} \quad (\text{not the LCB})$$

A declared deviation from optimism: the LP gets an **unbiased** spend estimate. Defended by ablation →

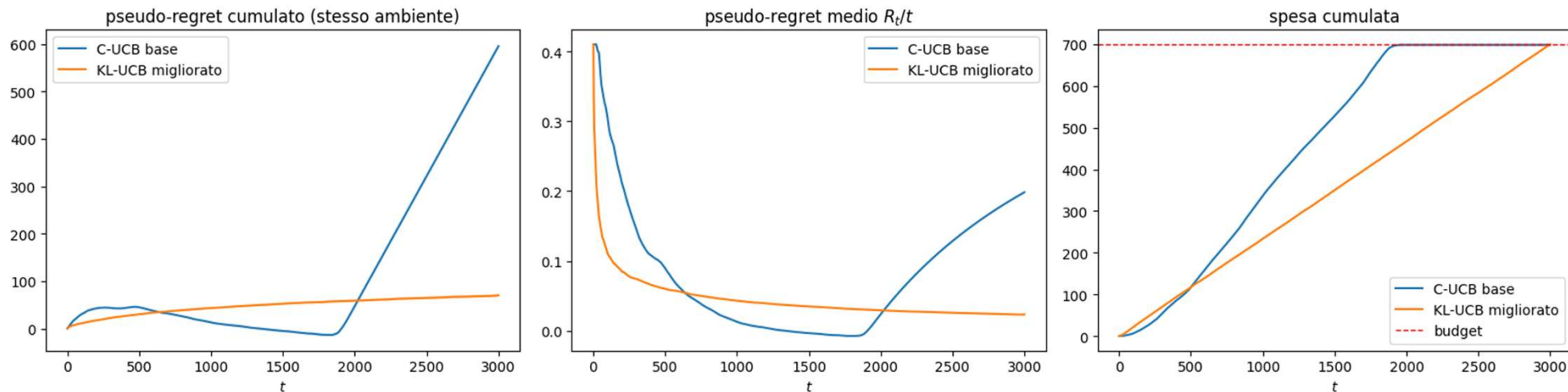
4 · DYNAMIC PACING

$$\rho_t = B_t / (T - t)$$

The LP cap tracks the remaining budget, restricted to still-affordable super-actions and corrects early spend drift.

Same LP structure as the course baseline; only the inputs change.

Results: Regret Stays Flat, Spend Reaches the Horizon



$\approx 595 \rightarrow \approx 70$

final pseudo-regret, same seeds

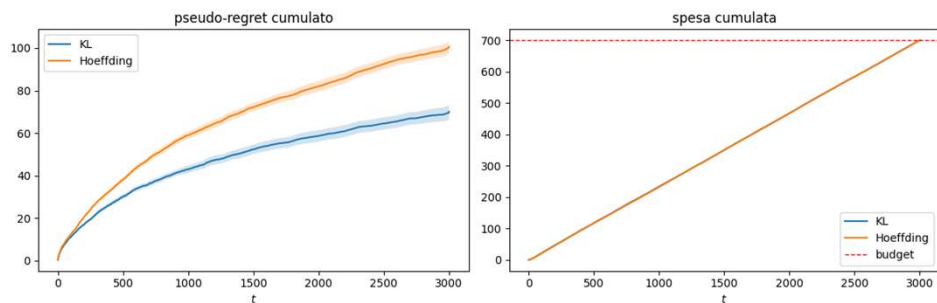
96%

of the oracle's expected reward

0.023

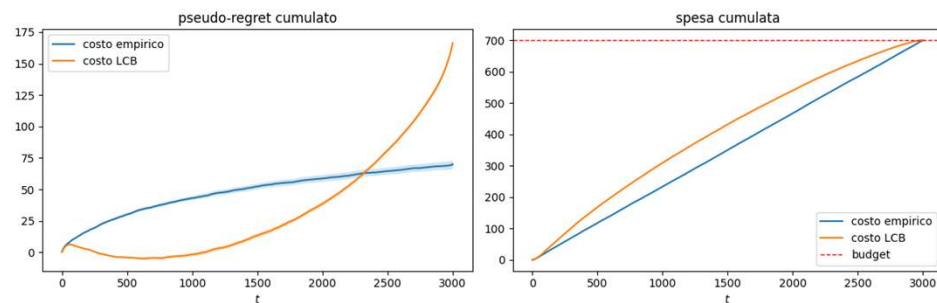
R_t/t at the horizon — still decreasing

Ablations: Confidence Bound and Cost Estimate



**70 vs
100**

KL vs Hoeffding — only the bound changes; KL orders small-probability arms correctly.

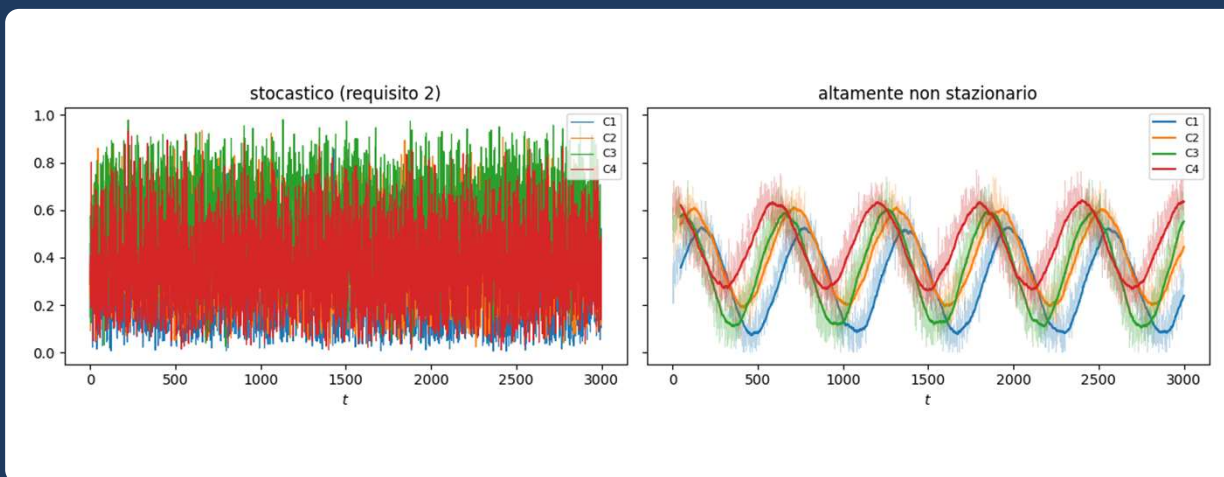


**70 vs
166**

Empirical vs LCB cost — LCB spends 241 vs 176 by $T/4$; the early advantage collapses at the end.

Best-of-Both-Worlds: the Primal-Dual Method

Same instance as Requirement 2, now with **full feedback** — after each round, thresholds reveal reward and cost of every super-action.



Stochastic: Beta marginals + common factor

Highly non-stationary: oblivious fast sinusoids + small noise

$T = 3000 \cdot B = 700 \cdot 250$ super-actions

Hard budget enforcement

BENCHMARK Best fixed distribution in hindsight satisfying the budget in expectation — we plot against full OPT, stricter than the p -OPT of the adversarial guarantee.

Baseline: Primal-Dual with Course Constants

PRIMAL — HEDGE ON 250 SUPER-ACTIONS

$$L_t(a) = f_t(a) - \lambda_t (c_t(a) - \rho)$$

DUAL — PROJECTED OGD, $\lambda \in [0, 1/\rho]$

$$\lambda_{t+1} = \Pi(\lambda_t + \eta (c_t \cdot x_t - \rho))$$

COURSE LEARNING RATES

$$\eta_H = \sqrt{\log K/T}, \quad \eta_{\text{OGD}} = 1/\sqrt{T}$$

Normalization choice: Hedge needs losses in $[0,1]$, but L_t contains a time-varying λ_t . Fixed worst-case bounds ($\lambda_{\max} = 1/\rho$) keep one scale for the whole horizon — at the price of a **range ≈ 11** , which divides Hedge's effective rate by the same factor, making it more conservative.

0.249 / 0.222

regret/T — stochastic / non-stationary (random: 0.436 / 0.391)

569 / 558

spend out of 700 — clear **under-spending**, no violations

Result

Primal-dual beats random in both regimes and never violates the budget, but converges slowly and leaves unused budget. The worst-case stop cannot explain the under-spend (it leaves at most one round of cost): it comes from the wide normalization and the generic constants.

Theory-Prescribed Learning Rates, Per-Action Budget Check

1 · RATES FROM THE THEORY

$$g_t(a) = f_t(a) - \lambda_t c_t(a)$$

$$\eta_H = \sqrt{8 \cdot \log K / T}, \quad \eta_{\text{OGD}} = \lambda_{\max} / (G\sqrt{T})$$

Hedge updates only on the action-dependent part of the Lagrangian ($\lambda_t p$ is constant over actions) and uses the same fixed normalization. Not a new method — uses constants prescribed by Hedge and OGD theory for these specific ranges.

2 · PER-ACTION BUDGET CHECK

a_t unaffordable $\Rightarrow$ play null action

Replaces the global worst-case stop: only the sampled action that cannot be paid is substituted. Controls the action we actually execute instead of the worst case.

Declared asymmetry: Hedge and the dual still update on the proposed x_t , while the budget decreases with the cost of the executed action.

Override frequency is ≈ 0 in both regimes.

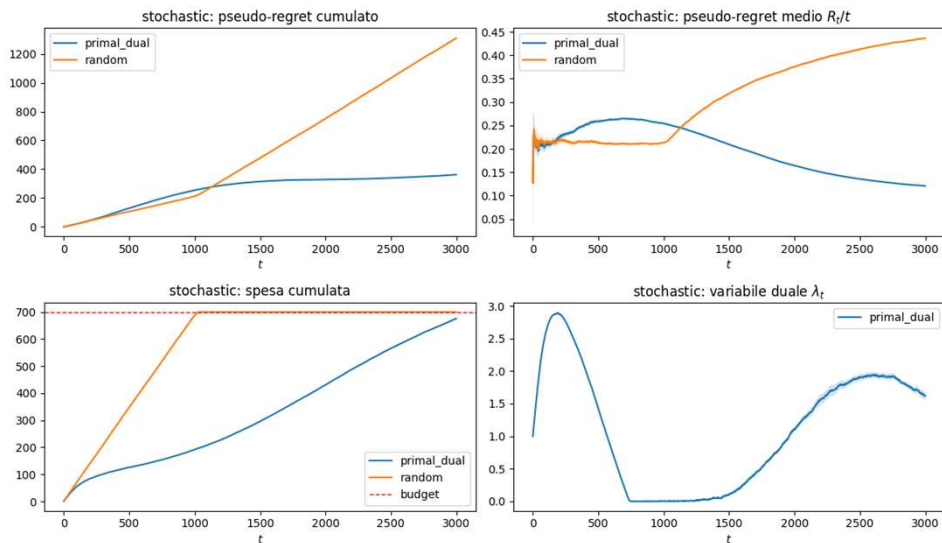
NO DYNAMIC PACING HERE

In the primal-dual, dynamic pacing is already encoded in the dual variable: λ_t rises when average spend exceeds p and depresses costly bids.

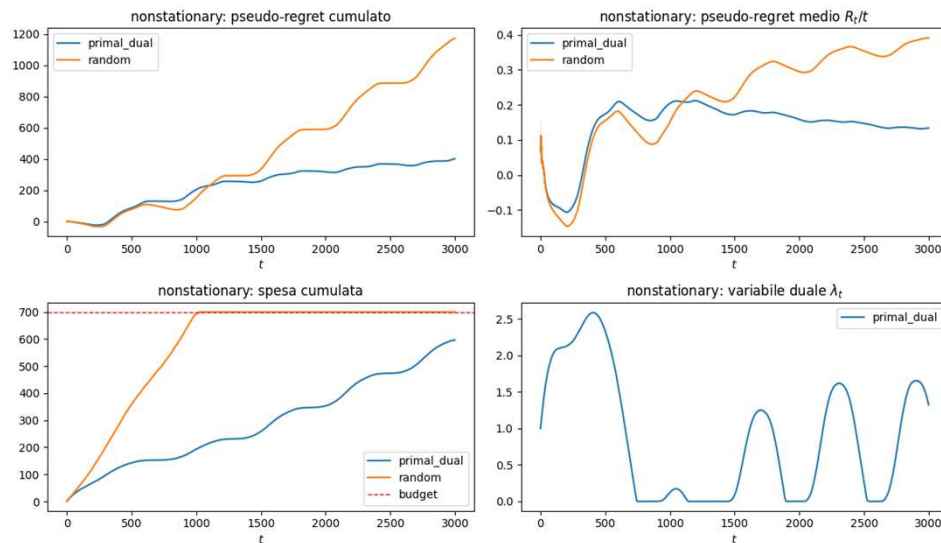
Ablation Study: The ablation study yields identical regret and expenditure for both rules, proving that performance gains stem entirely from optimal Hedge/OGD learning rates; however, the per-action constraint is preserved for its logical superiority in managing actual sampled actions over conservative worst-case shutdowns.

Results in Both Regimes

STOCHASTIC



HIGHLY NON-STATIONARY

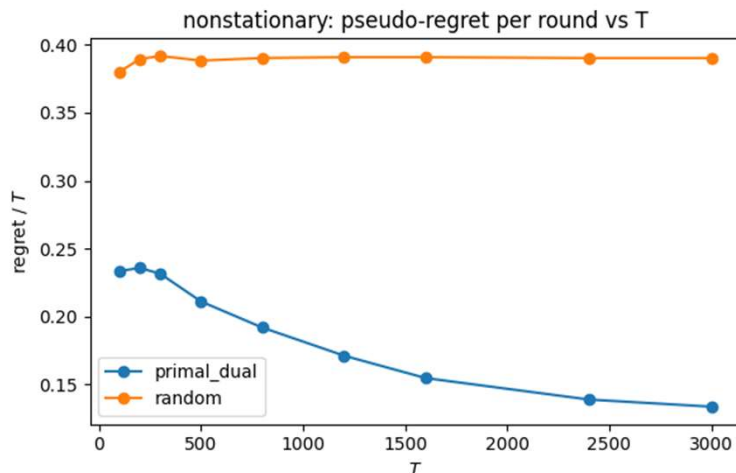
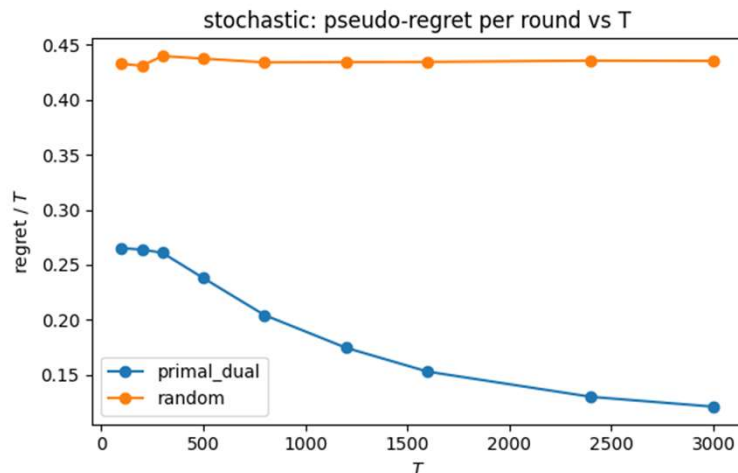


0.249 → 0.121 regret/T,
stochastic

0.222 → 0.134 regret/T, non-
stationary

569→676 • 558→597
spend out of 700 — never a violation

Horizon Sweep: Best-of-Both-Worlds



0.27 → 0.12

regret/ T as T grows, stochastic — consistent with sublinear regret

0.23 → 0.13

regret/ T as T grows, non-stationary — same algorithm, same constants

Expected best-of-both-worlds behavior: sweeping from 100 to 3000, regret/round of primal-dual decreases in both regimes, while the random stays high and almost flat.

Slightly Non-Stationary Environments with Multiple Campaigns

Combinatorial-UCB from Req 2, extended with a **sliding window** and a **change detector**, compared with the **primal-dual** method from Req 3. At each change point the competitor bid distributions change — and with them which campaigns are profitable.

HORIZON

T = 3000

3 equal segments, $L = 1000$, 2 known change points ($Y_T = 2$).

BUDGET

B = 560

First-price auctions, 4 campaigns, $p = B/T \approx 0.187$.

ALGORITHMS

SW Combinatorial-UCB

CUSUM Combinatorial-UCB

Primal-dual (full feedback)

BENCHMARK

≈ 1229

segment-local oracle: one LP per segment, same p per round.

REQUIREMENT 4 • SETUP

Environment

T = 3000 • 3 segments • B = 560

One bid per campaign each round, from a **grid of 8 values**, denser on low bids (step 0.10, then 0.15).

Conflict chain C1–C2–C3–C4: adjacent campaigns cannot both bid → **168 super-actions**, 27 positive arms.

Each campaign has a hidden threshold $m_i \sim \text{Beta}(\alpha_{s,i}, \beta_{s,i})$, fixed within a segment, new at each change point.

Win where $b_i \geq m_i$, pay the bid (first price): utility $v_i - b_i$, cost b_i .

All campaigns draw from a **shared budget** B = 560, $\rho = B/T \approx 0.187$ per round.

The cheapest campaign rotates at every change point:

SEGMENT 1
t = 1–1000
cheapest: **C1**



SEGMENT 2
t = 1001–2000
cheapest: **C2**



SEGMENT 3
t = 2001–3000
cheapest: **C3**



distributions of m_i · x: 0 to 1

— C1 — C2 — C3 — C4

Methods

all hyperparameters from course formulas, Υ_T known

FORGETS BY CONSTRUCTION

SW-UCB



- cache of the last W rounds only
- bonus computed with $\log W$
- arm unseen in window: one forced pull

$$W = 2\sqrt{T \log T / \Upsilon_T} \approx 219$$

DETECTS, THEN RESETS

CUSUM-UCB



- M reference samples per arm $\rightarrow$ mean
- statistic accumulates deviations; over $h \rightarrow$ arm reset
- α -exploration keeps arms monitored

$$M = 7, h \approx 14.6, \alpha \approx 0.07$$

NEVER FORGETS

Primal-dual

$$L(a) = f(a) - \lambda(c(a) - \rho)$$

- Hedge on $K = 168$ super-actions, full feedback
- OGD on the dual λ (budget price)
- gains normalized in $[0,1]$, per-action budget stop

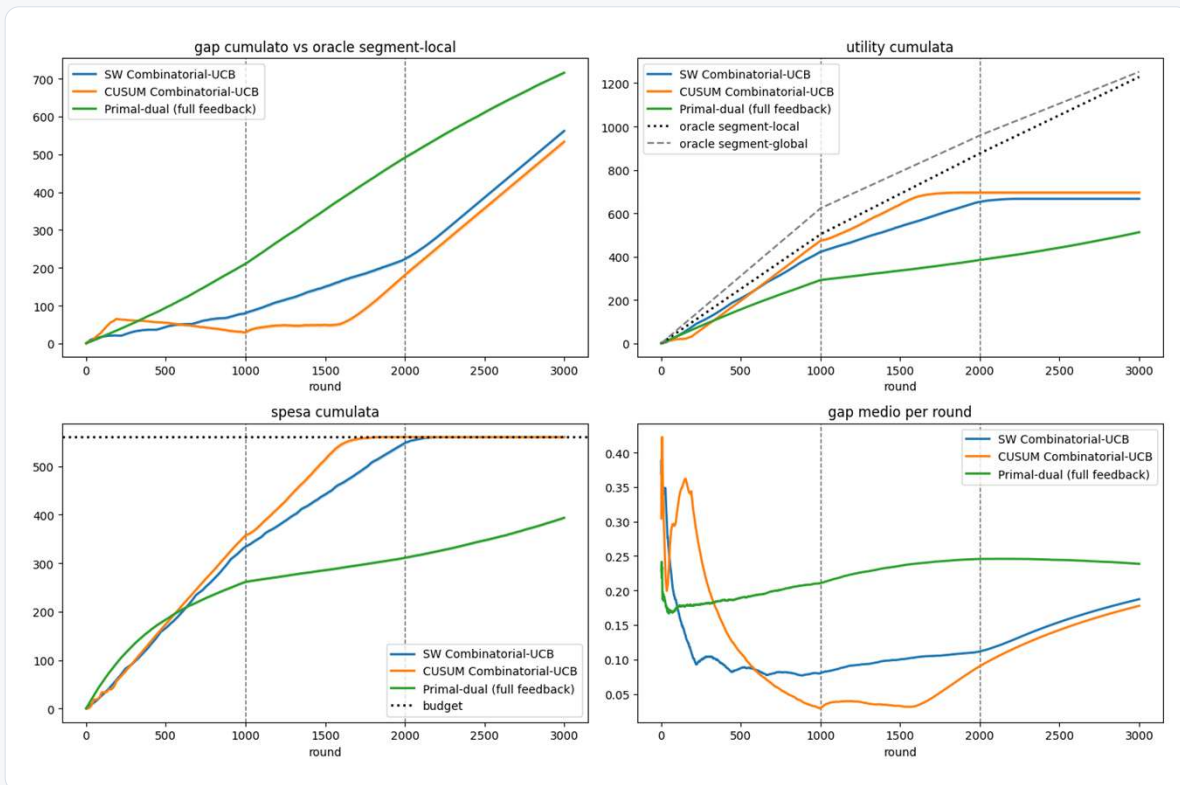
$$\eta_{\text{Hedge}} \approx 0.041, \eta_{\text{OGD}} \approx 0.018$$

SHARED DECISION LAYER (REQ 2)

Both UCBs feed reward UCB + cost LCB into an LP over the super-action simplex, cap $p = B/T$ fixed.

Baseline, Course Formulas

30 trials · final gap vs segment-local oracle ≈ 1229



Mean over 30 trials, shaded SEM · dashed lines: change points · dotted: oracles and budget.

SW-UCB

562

CUSUM-UCB

533

Primal-dual

716

PROBLEMS TO FIX

- ① both UCBs spend all of B, but **too early** : cost LCB understates costs, fixed p never corrects
- ② primal-dual leaves **167 of 560 unused** : infinite memory, conservative
- ③ CUSUM fires **~18 resets vs 54 reference** : too few samples per arm to reach $h \approx 14.6$ (no hand-tuning of h)

Improved Version: Three Declared Changes

same environment, same benchmarks, same seeds

01 · WIDER WINDOW

$$W = L/2 = 500$$

- half a segment: widest window that stays comfortably stationary
- more samples per arm in the combinatorial setting
- derived from known quantities, **not tuning**

ablation, 30 trials each: gap **496 → 417** vs $W = 219$

02 · EMPIRICAL COST

$$\hat{c}(a) = b \hat{p}$$

- replaces the cost LCB in the LP
- fixes the early budget exhaustion

supported by the Req 2 ablation: pseudo-regret **70 vs 166**

03 · DYNAMIC PACING

$$\rho_t = B_t / (T - t)$$

- LP cap = remaining budget / remaining rounds
- recovers spend drift around change points

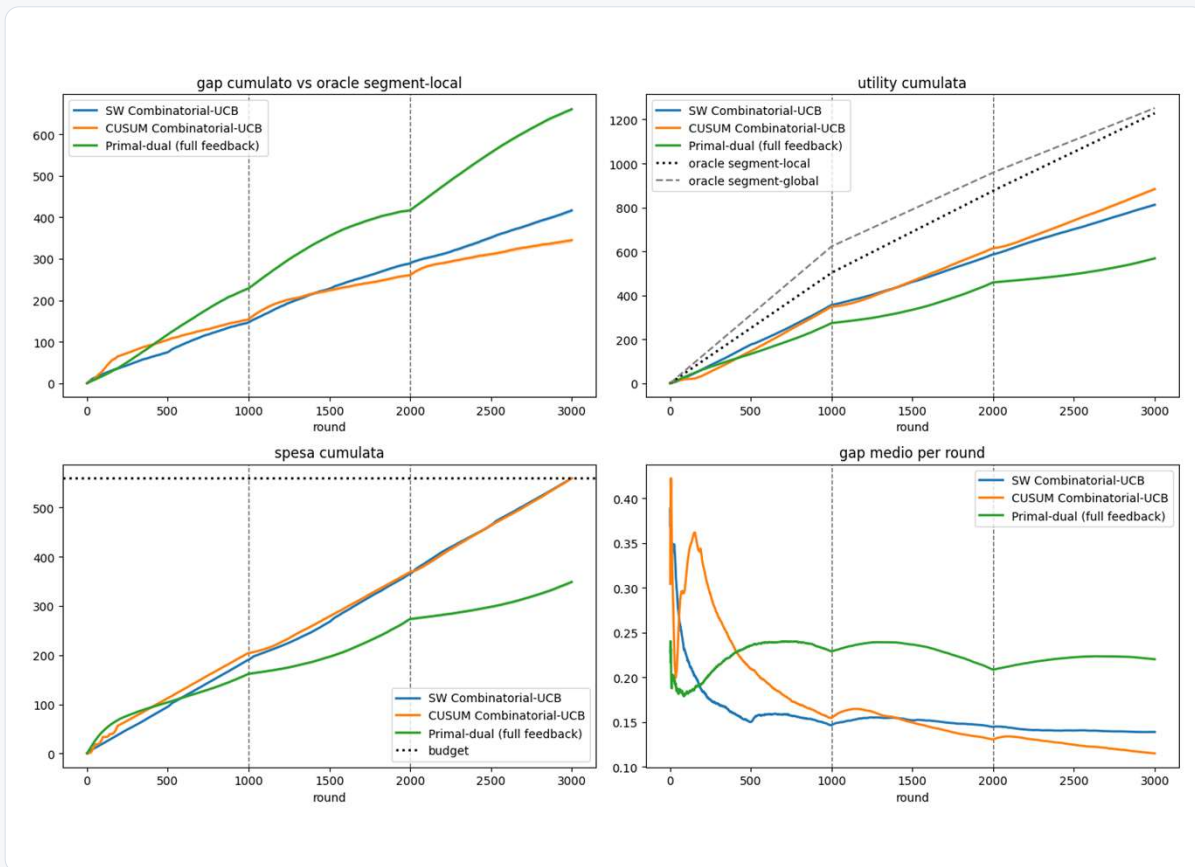
same modification used in **Req 2**

CUSUM UNCHANGED — detector keeps the course formulas; only the decision layer changes (02, 03).

PRIMAL-DUAL UPGRADED — Req 3 theoretical rates, so each method enters the comparison in its best version.

Results of the Improved Version

30 trials • same seeds • budget respected in every trial



FINAL GAP, BASELINE → IMPROVED

CUSUM-UCB 533 → 345

best method; resets concentrate near the true change points

SW-UCB 562 → 417

wider window, empirical cost, dynamic pacing

Primal-dual 716 → 660

theoretical rates; still leaves 211 of 560 unused

Light bar: baseline • solid bar: improved • scale up to 716.

Spend now **reaches the horizon** (pacing + empirical cost); the wider window fixes the sample side. CUSUM resets: **~18 → ~32** per trial.

Comparison and Limits

average gap per round, improved versions

0.115

CUSUM-UCB, best: active resets refresh stats after each change

0.139

SW-UCB: passive forgetting pays a fixed variance price

0.220

Primal-dual: 0.239 with lab constants

TAKEAWAYS

- non-stationary UCBs clearly beat the primal-dual; CUSUM slightly ahead of SW
- primal-dual chases the best **fixed** distribution: wrong target on three segments
- theoretical rates move it only $716 \rightarrow 660$, and 211 of 560 stay unused: the limit is **structural, not tuning**
- semi-windowed primal-dual tried and discarded: better than full, still below both UCBs, outside the course

LIMITS

- CUSUM with course formulas stays weakly reactive with many arms
- the segment-global oracle (moves budget across segments) is out of reach for every online learner
- the gap does not vanish: repeated exploration after each change grows with the number of arms



Thank You